

SQL Filtering: Login Activity and Employee Device Updates

Simulated Case Study

Project Overview

As part of a simulated security investigation, I used SQL to examine login activity and employee records stored in an organization's database. The work involved filtering authentication records to investigate potentially suspicious login activity and identifying employee devices that required security updates.

I worked with the `log_in_attempts` and `employees` tables to find the information needed for each task. SQL filters allowed me to narrow the results based on login times, dates, locations, and employee information without having to review every record individually.

Login Activity Review

Failed Login Attempts After Business Hours

I reviewed the `log_in_attempts` table to find unsuccessful login attempts after 18:00. By combining the time and login outcome with `AND`, I could narrow the results to failed attempts that occurred outside business hours.

```
select * from log_in_attempts where login_time > '18:00' and success = 0;
```

Result: 19 failed login attempts after 18:00.

Login Attempts on Specific Dates

I retrieved login activity from May 8 and May 9, 2022, to review authentication records surrounding a reported suspicious event. The `OR` operator includes records from either date, allowing both days to be reviewed together.

```
select * from log_in_attempts  
where login_date = '2022-05-09' OR login_date = '2022-05-08';
```

Result: 75 login attempts across the two dates.

Login Attempts Outside Mexico

I filtered login records to exclude attempts associated with Mexico. Because the `country` column contains both `MEX` and `MEXICO`, I used `NOT LIKE` with the `%` wildcard to exclude both values in a single condition.

```
select * from log_in_attempts where not country like 'MEX%';
```

Result: 144 login attempts outside Mexico.

Employee Device Update Review

Marketing Employees in the East Building

I used the `employees` table to identify Marketing employees whose devices needed security updates in the East building. Combining the department and office filters with `AND` narrowed the results to the correct employees, while `LIKE 'East%'` included all offices in that building.

```
select * from employees where department = 'Marketing' and office like 'East%';
```

Finance and Sales Employees

A separate security update was required for employees in Finance and Sales. I used the `OR` operator to retrieve records from either department, providing the employee and device information needed for the update.

```
select * from employees where department = 'Finance' or department = 'Sales';
```

Employees Outside of Information Technology

The final update had already been completed for Information Technology employees, so I queried the remaining departments. Using `NOT` excluded Information Technology records while retrieving employees whose machines still required the update.

```
select * from employees where not department = 'Information Technology';
```

Result: 161 employees outside the Information Technology department.

Outcome

Using SQL, I retrieved login and employee records to support two security tasks: investigating potentially suspicious login activity and identifying devices that required updates. Filtering the database allowed me to focus on the records relevant to each situation, including failed logins after business hours, activity on specific dates, and employees working in particular departments or locations.

Combining conditions with `AND` and `OR`, excluding records with `NOT`, and using `LIKE` to match patterns helped me retrieve the information needed without reviewing the entire database manually. These queries provided a more focused way to investigate login activity and organize security updates across the organization.